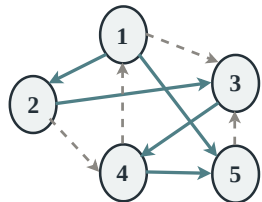


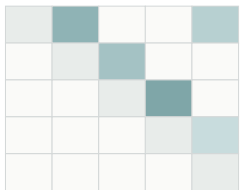
Inputs

Incomplete graph

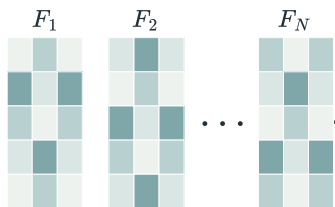


→ Observed edges
- - - Unobserved entries

Observed entries of A^*

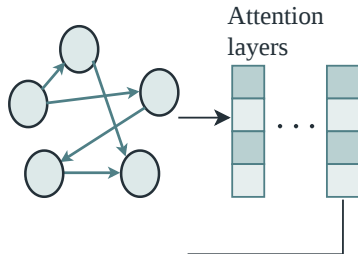


Node features F_i



GAT Edge Reconstruction

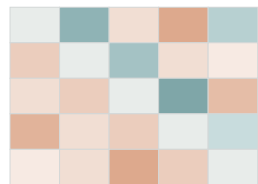
GAT encoder



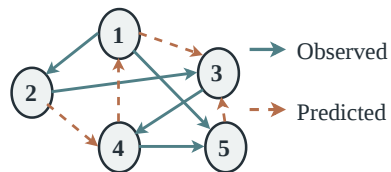
Edge decoder (MLP)



Completed adjacency $\tilde{A}$

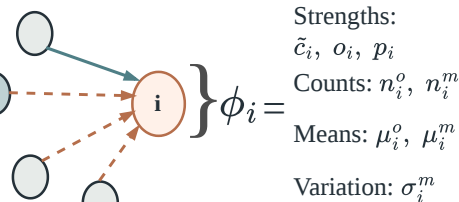


Completed graph



Strength-Guided Calibration

Incoming statistics (node i)



Standardize

$$z_i = (\phi_i - \mu_{dev}) \oslash s_{dev}$$

Residual calibrator

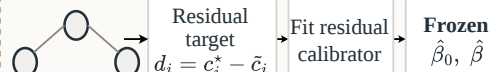
$$\hat{d}_i = \hat{\beta}_0 + \hat{\beta}^T z_i$$

Projection to calibrated in-strength

$$\hat{c}_i = \Pi_{[o_i, o_i + n_i^m]}(\tilde{c}_i + \hat{d}_i)$$

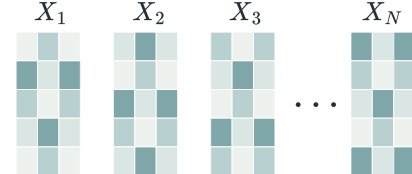
Offline calibration

Complete-reference
development graphs



Signal Fusion

Fusion signals X_i



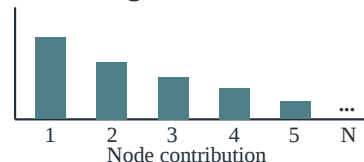
Each X_i : Samples $\times$ Channels

Normalize

$$\hat{w}_i = (\hat{c}_i + \varepsilon) / (\sum_k \hat{c}_k + N\varepsilon)$$

Fusion weights

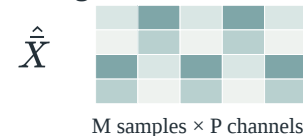
$$\sum_i \hat{w}_i = 1$$



Weighted fusion

$$\hat{X} = \sum_i \hat{w}_i X_i$$

Fused signal



M samples $\times$ P channels